

EFFECTS OF HYPERVENTILATION ON REPEATED PEDALING SPRINT PERFORMANCE: SHORT VS. LONG INTERVENTION DURATION

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ABSTRACT

Sakamoto, A, Naito, H, and Chow, C-M. Effects of hyperventilation on repeated pedaling sprint performance: short vs. long intervention duration. *J Strength Cond Res* 32(1): 170–180, 2018—Previously, hyperventilation (HV)-induced respiratory alkalosis, implemented during the last 30 seconds of each 60-second recovery that separated repeated pedaling sprints, has been shown to attenuate performance decrement. The present study investigated whether the ergogenic effects of HV would hold if the HV duration was shortened or extended. Seventeen power-trained athletes performed 10 seconds $\times$ 10 sets of standing pedaling sprints on a cycle ergometer, with a 60-second interset recovery and the load (kp) set at $0.075 \times$ body mass, under 3 breathing conditions: control, HV of 15 seconds (HV_{short}), and HV of 45 seconds (HV_{long}). Subjects breathed spontaneously during each 60-second recovery for the control condition. Under HV_{short} and HV_{long} conditions, subjects hyperventilated at 60 breaths $\cdot$ min⁻¹ with near-maximum tidal volume during the last 15 or 45 seconds, respectively, of each recovery period. Peak and mean pedaling power outputs (PO_{peak} and PO_{mean}) were documented for each sprint set to compare performance decrements between conditions. No significant condition effect or condition $\times$ time interaction was found for PO_{peak} and PO_{mean}. The lack of ergogenic effects with HV_{long} may be ascribed to a complex interaction between the positive (augmented buffering effects) and negative effects of HV (decreased aerobic energy metabolism and exaggerated discomfort sensation of increased ventilatory work). For HV_{short}, the implemented duration may have been too short to yield positive physiological effects. A practical implication is that ergogenic effects may be impaired when HV is too short or too long, with

the duration of around 30 seconds being the target for this exercise type.

KEY WORDS hypocapnia, respiratory alkalosis, power, fatigue, recovery

INTRODUCTION

Intermittent activities, such as interval training or high-intensity resistance training, comprise repeated short bouts of maximum or near-maximum muscle contractions with recovery periods between bouts. Maximizing the rate of interset recovery may enhance the quality of subsequent muscle contractions, particularly the power output (PO) (faster speed and greater force outputs), which could ultimately be translated into greater training effectiveness. A proposed major fatigue mechanism during intense intermittent exercise has been an excessive [H⁺] release from ATP hydrolysis via an increased rate of anaerobic glycolysis (10,16,26). The resulting fall in intramuscular pH impairs the energy supplying systems of PCr and anaerobic glycolysis (19,20,27,31) and the excitation-contraction coupling mechanisms (8,10,22,25). Moreover, cumulative anaerobic challenges that approach or exceed the tolerance limit may lead to central fatigue, a voluntary or autonomic reduction of central motor commands (1,13,14,16,23).

Voluntary hyperventilation (HV), incurring respiratory alkalosis, has theoretical rationales for attenuating the abovementioned fatigue effects. It accelerates the excretion of CO₂, drives the direction of equilibria of the bicarbonate buffering system equation toward removal of H⁺, and elevates pH in both intra- and extracellular environments at rest, under submaximal challenges, or after a single bout of maximum effort (3,4,21,32). These previous HV studies, however, demonstrated no ergogenic effects because the exercise modalities used were not suited to elicit an ergogenic response. Should the buffering effects be sustained during intense intermittent activities, then HV would enhance performance by countering acidity. Hyperventilation also increases the neuronal and muscle excitabilities at rest or prevents the deterioration of the

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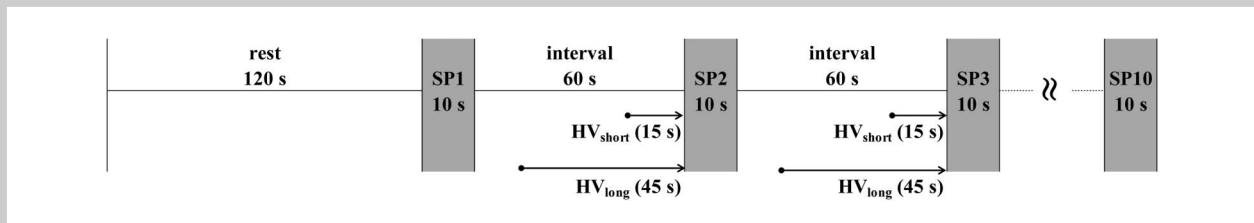


Figure 1. A flow diagram of experimental protocol. SP = sprint pedaling. For the control condition, subjects breathed spontaneously during each 60-second recovery period. Hyperventilation (HV) was implemented during the last 15 seconds of each 60-second recovery for HV short (HV_{short}) or the last 45 seconds for HV long (HV_{long}). Blood pH and PCO₂ were measured at rest (baseline) and then after the third, sixth, and ninth pedaling sets. Blood [La⁻] was measured at rest (baseline) and then after every second sprint sets.

excitabilities during intense exercise, i.e., stabilization of the M-wave (11,12,30), although these states may not necessarily influence performance (11). Accordingly, we implemented voluntary HV during the last 30- of 60-second interset recovery, which separated 10 sets $\times$ 10-second pedaling sprints, and showed an attenuation of performance decrement over repeated sprint sets (28). The proposed mechanisms for the ergogenic effects may include sustained rates of PCr and glycolytic energy supplies (4,7,15,20,27,31) and delayed impairment in the excitation-contraction coupling (8,10,22,25). Additionally, a voluntary increase in activation of the respiratory muscles may divert the attention from fatigue sensation or stimulate the reticular formation to restore arousal, resulting in averted central fatigue: a mechanism called “the Setchenov Phenomenon” (1,2,16). The presence of the Setchenov Phenomenon during voluntary HV, however, has not been systematically proven.

It should be noted, however, that detrimental effects may be anticipated when HV that intervenes in the recovery period is excessive. The negative physiological effects of HV on the brain and the peripheral musculature include (a) decreased O₂ perfusion because of decreased blood flow through vasoconstriction, (b) decreased diffusion gradient because of a left shift of the oxyhemoglobin dissociation curve, and (c) a slower rate of O₂ uptake when pyruvate dehydrogenase activity was slowed (3,6,7,24). Thus, HV can potentially impede O₂ and fuel delivery at a time when it is required to enable PCr resynthesis and the Cori cycle (18) during the recovery period. Excessive HV may also reduce the PO in the subsequent exercise bout by decreasing central motor commands through reduced cerebral blood flow (6,24). Hyperventilation that is shortened (e.g., less than 30 seconds), on the other hand, may not sufficiently yield alterations in blood gases that would serve to buffer the acidic muscle milieu.

Hence, this study aimed to examine the impact of HV duration on the performance outcome of repeated pedaling sprints using the same exercise protocol as our previous study (28), with the duration of HV implemented during the

inter-set recovery, however, being shorter (15 seconds) or longer (45 seconds) than our previous method (30 seconds).

METHODS

Experimental Approach to the Problem

This study compared the repeated sprint performance between 3 breathing conditions (control vs. HV short [HV_{short}] vs. HV long [HV_{long}]) in a crossover counterbalanced design. The 3 conditions were separated by 24–72 hours and completed within 7 days. The exercise protocol consisted of 10 sets of 10-second standing pedaling sprints on a cycle ergometer with a 60-second recovery between sets. For the control condition, subjects breathed spontaneously during the 60 seconds of each recovery. For the HV_{short} condition, after the first sprint set, subjects hyperventilated during the last 15 seconds of each recovery. For the HV_{long} condition, subjects hyperventilated during the last 45 seconds of each recovery. Peak and mean pedaling POs achieved within each sprint set were used to evaluate the performance decrement over 10 repeated sets. Respiratory parameters were monitored breath by breath during the exercise test, including respiratory rate (RR), expired tidal volume (VT), minute ventilation (VE), and end-tidal partial pressure of CO₂ (P_{ET}CO₂), to control for the methodological HV and to document altered respiratory patterns resulting from the HV interventions. Blood pH, PCO₂, and [La⁻] were analyzed to report changes in blood gas status and lactate kinetics associated with HV (greater [La⁻] was expected because of increased production or efflux of lactate and decreased lactate clearance) (3–5). The respiratory and blood parameters also served to evaluate physiological strains imposed during the exercise.

Subjects

Twenty-one power-trained university athletes were recruited for this study. Of these, 17 subjects (14 men and 3 women) with 8 cyclists, 3 throwing event athletes, 5 rugby players, and 1 judo player were competent to complete the practice test and all experimental procedures. The mean age, height, and body mass of the 17 subjects were Mean $\pm$ SD 20.3 $\pm$ 0.9 years (range: 19–22 years), 171.9 $\pm$ 6.6 cm, and 72.2 $\pm$

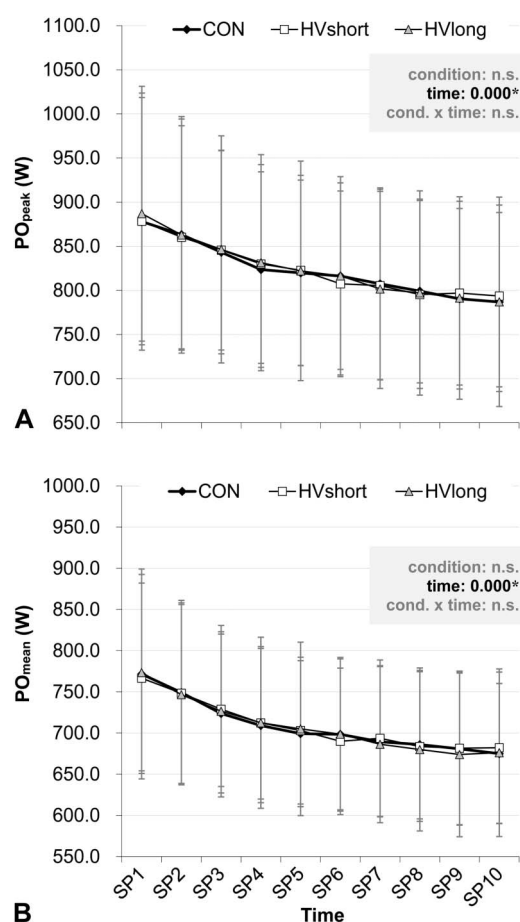


Figure 2. Changes in peak power output (PO_{peak}) (A) and mean power output (PO_{mean}) (B) over 10 repeated sprint sets comparing CON vs. Hyperventilation (HV) short (HV_{short}) vs. HV long (HV_{long}) conditions ($n = 17$). *Significant main effect according to 2-way repeated measures analysis of variance (condition $\times$ time). n.s. = statistically nonsignificant.

8.8 kg, respectively. They had competed in their sports for 6.8 ± 2.8 years and trained 5.5 ± 0.7 days per week at the time of the experiment. Before the experiment, all subjects reported to the laboratory on 2 separate occasions. On the first nonexperimental day, subjects were informed of the experimental procedures and any risks associated with the experiment by means of verbal instructions and written guidelines and gave informed written consent. On the second day, subjects practiced the pedaling sprints and HV methods used in this study. In the first part of the practice session, subjects practiced the HV_{short} and HV_{long} methods during the recovery periods involving 5 pedaling sets. The first pedaling set was performed maximally, followed by 4 pedaling sets at submaximal effort (70–80% first set) to avoid total exhaustion. During the second part of the practice session, subjects made a full practice run with 10 sets of repeated pedaling sprints at maximal effort with 3 bouts each

of control breathing, HV_{short} or HV_{long} during the recovery period between sets; each breathing method was self-selected or randomly allocated. This practice run was used to assess the subjects' capabilities to complete the exhaustive exercise. Those who were incapable of completing the second part of the practice because of severe exhaustion or nausea were excluded from the experiment ($n = 4$). On the practice day and the subsequent experimental days, subjects were instructed to eat a carbohydrate-rich meal 2 hours before the exercise test for fuel repletion and encouraged to drink plenty of water to avoid dehydration. No subjects consumed caffeinated or alcoholic drinks during the experimental period, and none were smokers. This study was approved by the Human Ethics Committee of Juntendo University, Graduate School of Health and Sports Science.

Procedures

Warm-up. Upon arrival in the laboratory on the experimental days, subjects warmed up on a cycle ergometer (Powermax-VII; Combi Wellness, Tokyo, Japan), a continuous pedaling at their preferred cadence for 10 minutes under 2.0 kp for men and 1.5 kp for women. The pedaling load was reduced by 0.5 kp if subjects felt it too heavy or fatiguing. After that, the pedaling load was set at 2.0 kp for all subjects, and they practiced a brief (3–5 seconds) explosive standing sprint every minute 5–7 times. During the recovery interval between sets, the breathing method assigned on that day was also practiced with correct timing of HV onset at a respiratory rate of 60 breaths $\cdot$ min⁻¹. Subjects consumed 500–1,000 ml of standard mineral water during or after the warm-up to maintain euhydration before the exercise test.

Repeated Sprints. The exercise test consisted of a 2-minute rest on the saddle followed by 10 seconds $\times$ 10 sets of maximal standing pedaling on the aforementioned cycle ergometer, with a 60-second recovery between sets (Figure 1). The saddle and handle heights were determined during the nonexperimental practice session and kept constant throughout the experiment. To determine the pedaling load ($0.075 \times$ body mass), the body mass was measured on the practice day and once again on the first experimental day. The pedaling load determined on the first experimental day was again used for the remaining experimental days. During the initial 2-minute rest and each recovery period, subjects remained seated with minimal or no leg movements. Five seconds before each sprint, subjects took a standing ready position and were counted down, and upon the "go" signal, sprint pedaling began with maximal effort. No verbal encouragement was given to the subjects during each sprint to establish a self-motivated state, a more ideal setting to allow the development of central fatigue and the counter effect of the Setchenov Phenomenon expected from HV (2,16,28,29). The ergometer was connected to a personal computer with the auxiliary software (Powermax VII; Combi Wellness) to record pedaling POs at 10 Hz. During later data

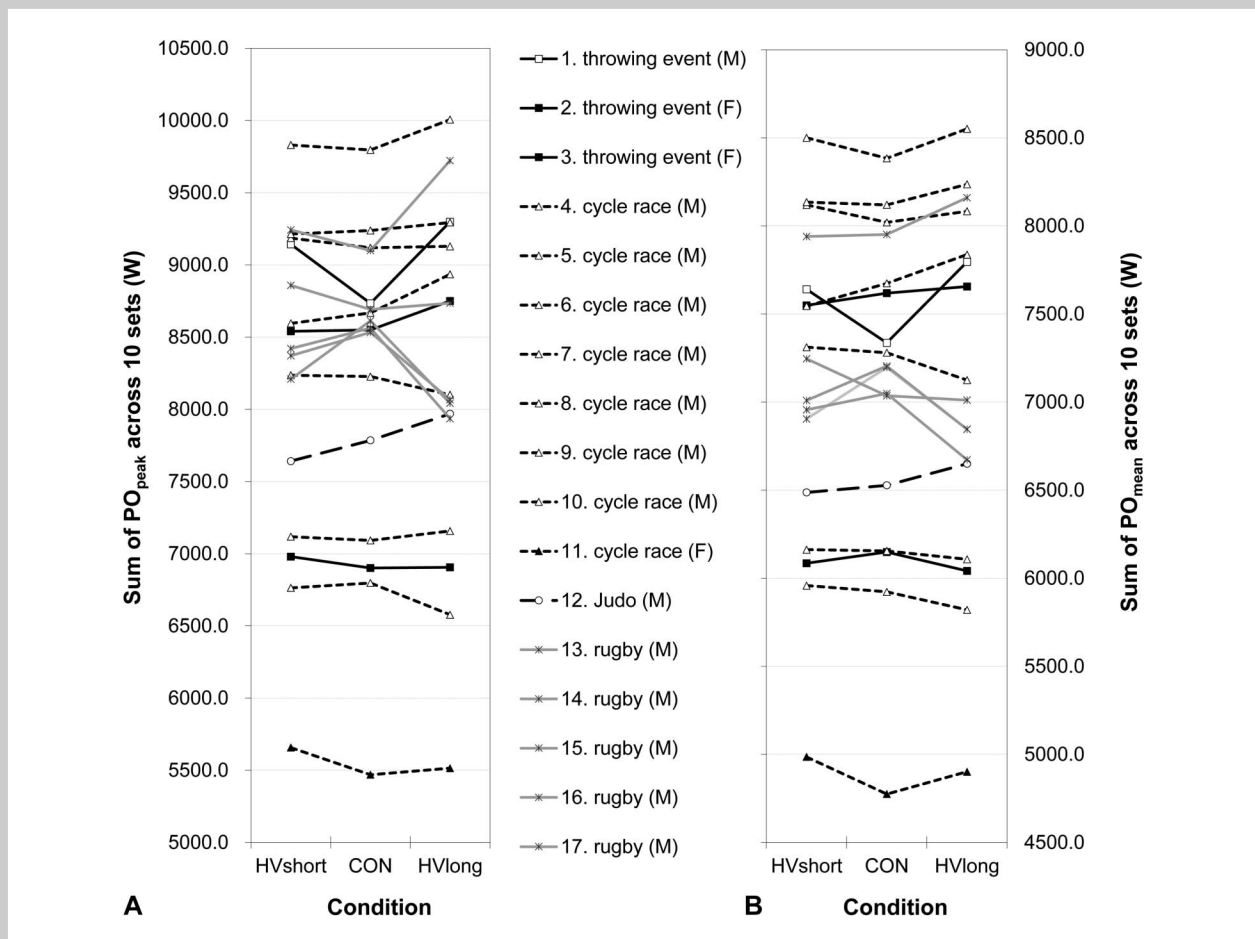


Figure 3. V frame displays of the sum of peak power output (PO_{peak}) (A) and sum of mean power output (PO_{mean}) (B) across 10 sprint sets per individual ($n = 17$) with different types of sport, sex, and power abilities; hyperventilation (HV) short (HV_{short}) and HV long (HV_{long}) data were plotted on either side of control data. An upright V frame displays ergogenic effects of HV_{short} and HV_{long}.

processing, the highest PO (PO_{peak}) and the average value (PO_{mean}) were identified within each sprint set. Their decrements over 10 repeated sets were used to evaluate changes in performance. All tests were conducted in a temperature- and humidity-controlled chamber at 20° C and 60%, respectively. To minimize diurnal variations in performance across the 3 experimental days, the exercise tests were carried out at similar time of day, with the maximum time difference being 3 hours, and completed within 7 days (24–72 hours apart between experimental sessions). To avoid fatigue effects, the experiments were scheduled before their own training sessions, which were, however, not feasible for some subjects because of their frequent training routines and university curricula. In these cases, the same conditions were instituted for the 3 testing days by having these subjects undergo the exercise test always after their routine training.

Breathing Conditions (CON vs. Hyperventilation Short vs. Hyperventilation Long). The first sprint set was initiated from

normal resting breathing for all the 3 conditions. For the control condition (CON), subjects continued spontaneous breathing during each of the recovery periods after the first sprint set. For the HV_{short} condition, subjects breathed spontaneously during the first 45 seconds of each recovery, followed by 15 seconds of HV (respiratory rate set at 60 breaths·min⁻¹ with near-maximum VT). For the HV_{long} condition, subjects breathed spontaneously during the first 15 seconds, followed by 45 seconds of HV (Figure 1). Sudden changes in breathing patterns because of puffing or unconscious deep breaths were avoided especially for the control condition. Subjects breathed freely during the sprint bouts.

The ventilatory maneuver chosen for the HV_{short} and HV_{long} conditions, i.e., near-maximal VT × 60 breaths·min⁻¹, was to replicate our previous breathing method of 30 seconds, which aimed to achieve the target P_{ET}CO₂ values between 20 and 25 mm Hg (28). Thus, the same maneuver was achieved in both the current and our previous studies, except that P_{ET}CO₂ was a dependent variable in this study.

TABLE 1. The respiratory parameters before each sprint set with or without HV ($n = 15$, 2 subjects were excluded from the data set because of a transducer error).*†

Variables	Condition	Pre-SP1 (baseline)	Pre- SP2	Pre- SP3	Pre- SP4	Pre- SP5	Pre- SP6	Pre- SP7	Pre- SP8	Pre- SP9	Pre- SP10	p for the main effects and interactions
RR (breaths·min ⁻¹)	CON	21.6	31.2	34.6	38.0	38.1	39.3	40.5	40.7	40.4	44.5	Condition: 0.000‡ (CON < HV _{short} = HV _{long})
	<i>SD</i>	4.3	6.8	4.7	6.4	6.2	7.2	7.7	7.6	8.5	9.2	
	HV _{short}	21.6	60.7	60.3	60.5	60.6	60.5	60.5	60.3	60.5	60.3	Time: 0.000‡
	<i>SD</i>	3.9	1.0	0.9	0.5	0.6	1.1	1.1	0.8	0.7	0.9	
	HV _{long}	23.6	60.5	60.1	60.6	60.7	60.5	60.5	60.2	60.1	60.3	Condition × time: 0.000‡
VT (ml)	<i>SD</i>	5.7	0.7	0.9	0.9	1.0	0.6	0.7	0.4	0.7	0.7	
	CON	1,004.6	1,984.7	2,155.7	2,178.5	2,181.3	2,120.3	2,096.7	2,161.9	2,036.7	1,984.3	Condition: 0.000‡ (CON < HV _{long} < HV _{short})
	<i>SD</i>	227.1	460.3	424.6	440.8	392.7	411.6	410.7	463.7	445.3	456.5	
	HV _{short}	974.5	2,500.6	2,537.8	2,514.5	2,554.6	2,560.7	2,545.6	2,531.6	2,504.5	2,501.6	Time: 0.000‡
	<i>SD</i>	228.0	466.6	439.0	407.0	454.4	453.6	442.7	431.1	435.5	440.6	
VE (L·min ⁻¹)	HV _{long}	937.1	2,310.5	2,340.3	2,341.6	2,385.6	2,358.9	2,307.6	2,272.0	2,288.0	2,309.7	Condition × time: 0.000‡
	<i>SD</i>	213.3	430.6	412.6	399.1	425.9	432.9	408.2	436.1	412.5	442.9	
	CON	20.5	60.4	73.6	81.8	82.4	81.9	83.2	86.4	81.1	85.9	Condition: 0.000‡ (CON < HV _{long} < HV _{short})
	<i>SD</i>	5.2	15.3	16.6	19.8	18.1	18.9	19.4	19.9	20.8	19.0	
	HV _{short}	20.1	151.5	153.1	151.9	154.6	154.9	154.0	152.4	151.3	150.7	Time: 0.000‡
P _{ET} CO ₂ (mm Hg)	<i>SD</i>	4.9	28.2	27.0	24.5	27.4	28.0	26.9	25.7	26.5	26.9	
	HV _{long}	20.4	139.7	140.9	141.6	144.7	142.4	139.1	137.0	136.8	139.2	Condition × time: 0.000‡
	<i>SD</i>	5.0	25.9	25.3	24.8	26.0	25.9	24.6	26.2	24.7	26.6	
	CON	39.5	44.7	42.6	40.1	39.0	37.1	35.8	35.1	35.1	34.0	Condition: 0.000‡ (CON > HV _{short} = HV _{long})
	<i>SD</i>	2.2	3.4	3.6	4.2	4.4	4.3	4.2	4.7	4.7	4.5	
$\dot{V}O_2$ (ml·min ⁻¹)	HV _{short}	38.3	27.1	27.8	27.1	26.4	25.4	24.8	24.2	23.8	23.7	Time: 0.000‡
	<i>SD</i>	1.8	2.2	2.4	2.4	2.6	2.4	2.6	2.4	2.4	2.7	
	HV _{long}	38.1	27.3	27.9	26.9	25.8	25.4	24.6	24.4	24.4	23.6	Condition × time: 0.000‡
	<i>SD</i>	2.8	2.1	2.4	2.5	2.7	2.6	2.6	2.7	2.4	2.0	
	CON	575.0	1,692.0	2,025.1	2,118.2	2,200.0	2,191.4	2,202.4	2,220.9	2,209.3	2,190.3	Condition: 0.008‡ (CON < HV _{long})

SD	105.3	218.8	307.8	322.3	335.6	328.9	327.4	330.1	342.8	330.9
HV _{short}	584.4	1,767.5	2,044.9	2,106.1	2,192.7	2,216.1	2,228.1	2,250.4	2,244.4	2,260.6
SD	102.5	214.6	243.2	260.8	276.1	279.1	289.6	289.2	288.8	302.5
HV _{long}	570.4	1,867.1	2,131.0	2,208.7	2,270.0	2,304.9	2,308.0	2,317.7	2,290.6	2,283.3
SD	95.8	252.3	301.6	310.1	314.2	313.2	306.5	318.6	321.4	313.2

Time: 0.000†
Condition × time:
0.000‡

*SP = sprint pedaling; RR = respiratory rate; HV = hyperventilation; HV_{short} = HV short; HV_{long} = HV long; VT = tidal volume; VE = minute ventilation; P_{ET}CO₂ = end-tidal partial pressure of CO₂.
†RR, VT, VE, and P_{ET}CO₂ were averaged over the last 30 seconds before the first sprint set (pre-SP1) and then the last 5 seconds before the 2nd–10th sets (pre-SP2 to pre-SP10). Vo₂ was averaged over the last 60 seconds before the first sprint set and then over the entire 60-second recovery period thereafter.
‡Statistical significance.

For the HV_{short}, we confirmed a significant alteration in the resting values of blood pH and PCO₂ (pH: 7.403 ± 0.014 → 7.471 ± 0.037, PCO₂: 38.3 ± 2.6 → 31.1 ± 3.6 mm Hg; $P < 0.001$, $n = 17$). Resting blood gases with HV_{long} were not investigated to avoid severe paresthesia or dizziness.

Respiratory Monitoring. Expired gases were sampled breath by breath through a fitted facial mask for RR, VT, VE, P_{ET}CO₂, and Vo₂ using an aeromonitor (AE-310S; Minato Medical Science, Osaka, Japan). The airflow sensor was calibrated using a 2-L syringe and O₂ and CO₂ sensors with known O₂ and CO₂ gas concentrations in accordance with the manufacturer's instructions before each experiment. The RR was controlled at 60 breaths·min⁻¹ for the HV_{short} and HV_{long} conditions using a metronome (set at 120 b·min⁻¹; inspiration or expiration per beat). Expired VT served to monitor the subjects' breathing efforts. Verbal feedback (e. g., breathe deeper) was given when a reduction in VT was detected during the HV intervention.

Breath-by-breath data of RR, VT, VE, and P_{ET}CO₂ recorded 5 seconds before each sprint set were averaged and used for statistical analyses, except for the resting baselines before the first sprint set. The baseline values of these respiratory parameters were established by averaging the values for the last 30 seconds of the rest period (RR at rest was slow). Vo₂ data were averaged for 60 seconds before the first sprint (baseline) and then for the entire recovery duration (60 seconds) to document changes in systemic O₂ uptake with different breathing conditions.

Blood Samples. Small amounts of whole capillary blood were collected from hyperemized earlobes into heparinized capillary tubes to measure blood pH, PCO₂, and lactate concentration ([La⁻]). From the left earlobe, approximately 60 µl of blood was collected at rest (baseline) and then immediately after the third, sixth, and ninth sprint sets for the measurement of blood pH and PCO₂ using a blood gas analyzer (Cobas b221 System; Roche Diagnostics, Tokyo, Japan). For [La⁻], blood was collected from the contralateral earlobe (20 µl) at rest (baseline) and then immediately after every second sprint set (2nd, 4th, 6th, 8th, and 10th sprints). The samples were analyzed using a lactate analyzer (Biosen S-Line; EKF Diagnostics, Barleben, Germany).

We did not collect blood samples immediately before any designated sprint sets (i.e., during the HV intervention), which could inform the levels of blood pH (alkalosis) and PCO₂ (hypocapnia) immediately after HV. However, it was not feasible to always collect sufficient amount of blood, especially for the HV_{short} intervention, because a 60-µl capillary tube required 10–30 seconds to fill.

Statistical Analyses

Statistical tests were performed using SPSS version 17.0 software (SPSS Inc., Chicago, IL, USA). The test-retest reliabilities of PO_{peak} and PO_{mean} between the 3 conditions

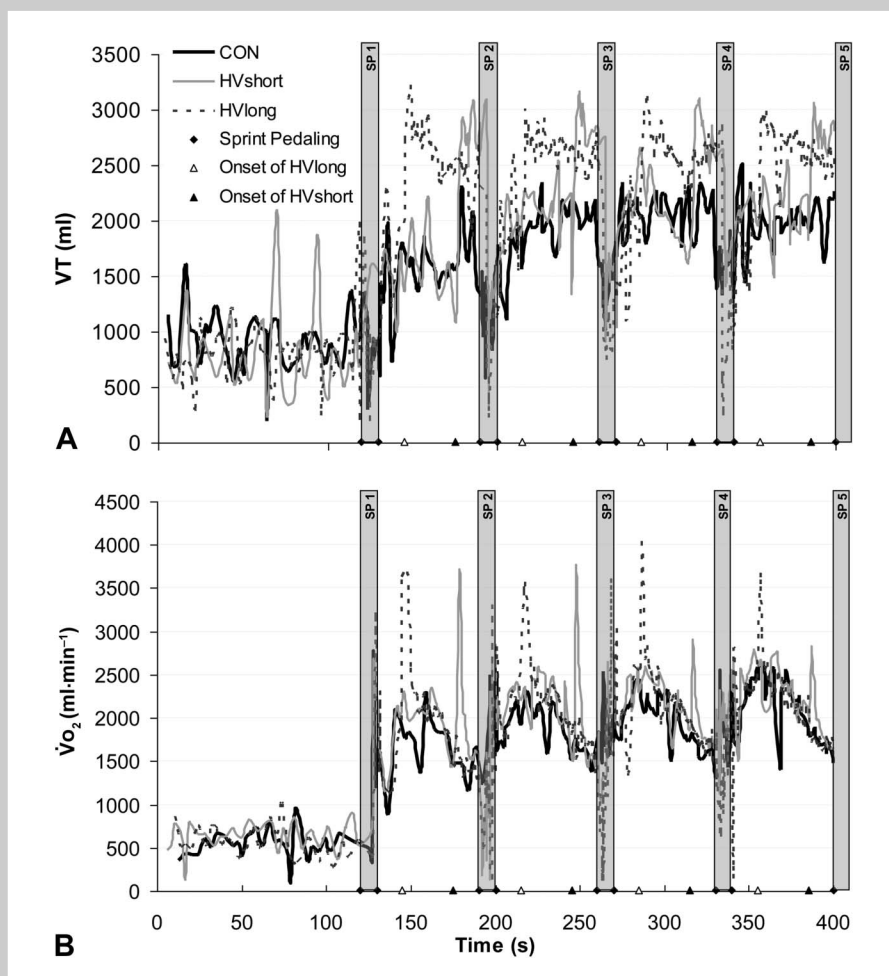


Figure 4. Typical breath-by-breath raw data of tidal volume (A) and $\dot{V}O_2$ (B) for the 3 breathing conditions up to the fifth sprint set, after which the data followed similar patterns. ◆ Start and end of pedaling sprint, △ start of hyperventilation (HV) long (HV_{long}), ▲ start of HV short (HV_{short}).

were evaluated by means of intraclass correlations (ICCs) using the values recorded during the first sprint set. The ICCs of PO_{peak} and PO_{mean} were, respectively, 0.982 and 0.969 (95% confidence interval: 0.960–0.993 and 0.932–0.987). One-way repeated measures analyses of variance (ANOVAs) were used to ensure the similarity in the baseline values of PO_{peak} and PO_{mean} , blood parameters (pH, PCO_2 , and $[La^-]$), and respiratory parameters (RR, VT, VE, P_{ETCO_2} , and $\dot{V}O_2$) between the 3 conditions. The main effect of breathing conditions (CON vs. HV_{short} vs. HV_{long}) and time (sprint set) and their interactions on PO_{peak} and PO_{mean} and the respiratory parameters (RR, VT, VE, P_{ETCO_2} , and $\dot{V}O_2$) were studied using 2-way repeated measures ANOVAs. For those cases violating the assumption of sphericity, significance was corrected using the Greenhouse-Geisser adjustment. When significant main effects were found, pairwise comparisons were performed where necessary using the Bonferroni method.

A complete data set could not be obtained from 2 subjects for the respiratory parameters because of a transducer error. Hence, the respiratory data of the remaining 15 subjects were used for statistical analyses.

The measurements of the blood gases were occasionally unsuccessful because of (a) unexpectedly long time taken for the self-cleaning process of the device resulting in clotting of the pending blood sample or (b) unidentified device errors. Hence, linear mixed model analyses were applied to blood pH and PCO_2 data for the main effects of condition and time and their interactions.

RESULTS

The PO_{peak} and PO_{mean} recorded during the first sprint set and the baselines of all other tested variables did not differ significantly between the 3 conditions.

Peak and Mean Power Outputs

Both PO_{peak} and PO_{mean} decreased significantly with sprint set ($P < 0.001$, Figure 2). No significant condition effect or condition $\times$ time interaction was, however,

observed for these variables (Figure 2).

The sum of power was calculated by adding the PO data over 10 sets for each subject (PO_{peak} and PO_{mean} separately) within each breathing condition. Figure 3 illustrates the sum of PO_{peak} (panel A) and the sum of PO_{mean} (panel B) identifying the sex, type of sport, and power abilities of the subjects. In these panels, the control condition is presented in the middle with HV_{short} and HV_{long} presented on either side, where the normal V frame reflects improved performance resulting from both HV methods. An inverted V frame indicates negative effects. The HV strategies produced inconsistent effects on performance resulting in numerous flat lines and both upright and inverted V frames (Figure 3).

Respiratory Parameters (Respiratory Rate, VT, Minute Ventilation, end-tidal partial pressure of CO_2 , and $\dot{V}O_2$)

The main effects of condition and time and their interactions were all significant for all respiratory parameters (Table 1).

TABLE 2. The blood parameters after designated sprint sets with or without HV ($n = 17$).^{*†}

Variables	Condition	Baseline (rest)	Post-SP1	Post-SP2	Post-SP3	Post-SP4	Post-SP5	Post-SP6	Post-SP7	Post-SP8	Post-SP9	Post-SP10	p for the main effects and interactions
pH	CON	7.402	—	—	7.297	—	—	7.256	—	—	7.241	—	Condition: NS
	SD	0.016			0.034			0.044			0.054		
	HV _{short}	7.403	—	—	7.293	—	—	7.254	—	—	7.234	—	Time: 0.000 [‡]
	SD	0.014			0.031			0.043			0.060		
	HV _{long}	7.403	—	—	7.307	—	—	7.247	—	—	7.234	—	Condition × time: NS
	SD	0.012			0.023			0.029			0.052		
PCO ₂ (mm Hg)	CON	39.3	—	—	38.5	—	—	34.4	—	—	32.2	—	Condition: 0.000 [‡] (CON > HV _{short} = HV _{long})
	SD	1.8			4.2			3.7			2.9		
	HV _{short}	38.3	—	—	37.0	—	—	31.3	—	—	29.1	—	Time: 0.000 [‡]
	SD	2.6			3.4			4.3			2.4		
	HV _{long}	38.4	—	—	35.0	—	—	30.4	—	—	26.8	—	Condition × time: 0.024 [‡]
	SD	2.5			2.8			3.6			3.5		
[La ⁻] (mM)	CON	1.8	—	4.9	—	8.7	—	10.5	—	11.6	—	12.1	Condition: NS
	SD	0.9		1.3		2.1		2.4		2.9		3.0	
	HV _{short}	1.7	—	4.7	—	8.5	—	10.6	—	11.9	—	12.6	Time: 0.000 [‡]
	SD	0.7		1.5		2.3		2.6		2.9		3.1	
	HV _{long}	1.4	—	4.8	—	8.9	—	11.1	—	12.5	—	13.1	Condition × time: 0.034 [‡]
	SD	0.6		1.2		1.8		1.9		2.2		2.4	

^{*}SP = sprint pedaling; HV = hyperventilation; HV_{short} = HV short; HV_{long} = HV long; NS = statistically nonsignificant.

[†]Blood pH and PCO₂ were measured at rest (baseline) and then immediately after the third, sixth, and ninth sprint sets (post-SP3, SP6, and SP9). Blood [La⁻] was measured at rest and then immediately after every second sprint set (post-SP2, SP4, SP6, SP8, and SP10).

[‡]Statistical significance.

Overall, both the HV conditions resulted in greater RR, VT, and VE and lower P_{ETCO_2} than the control condition, with the target constant RR (60 breaths·min⁻¹) well achieved (Table 1). These values were more stable over 10 sprint sets for the 2 HV conditions because of the respiratory manipulation compared with the controls. By contrast, the control condition exhibited progressive increases in RR, VT, and VE in early exercise sets followed by relatively steady values thereafter and a progressive decrease in P_{ETCO_2} throughout (Table 1).

Pairwise comparisons, however, revealed that the VT values were lower, as were VE, for the HV_{long} condition than HV_{short} ($P < 0.001$, Table 1). Consequently, the P_{ETCO_2} values did not differ between the HV_{short} and HV_{long} conditions despite different implementation durations (Table 1). Figure 4A shows the typical raw data of VT for the 3 conditions up to the fifth sprint set. VT was elevated at the onset of each HV_{short} and HV_{long} intervention. However, a gradual fall in VT was observed throughout the intervention period in the HV_{long} condition, explaining the lower VT values of the last 5-second recovery than those of the HV_{short} condition (Table 1).

$\dot{V}O_2$ values during the interset recovery were significantly affected by the conditions (Table 1, $P < 0.008$). According to the pairwise comparisons, the condition effect, however, was evident between the control and HV_{long} conditions only ($P < 0.016$), with greater $\dot{V}O_2$ for the HV_{long} condition. A significant condition $\times$ time interaction ($P < 0.002$) implied that for some interval sets, the $\dot{V}O_2$ differences between these 2 conditions were reduced (Table 1). Figure 4B revealed that $\dot{V}O_2$ increased immediately after the onset of both HV_{short} and HV_{long} interventions, however, followed by an abrupt fall. This prompt returning of $\dot{V}O_2$ explains the similar $\dot{V}O_2$ between the control and HV_{short} conditions and reduced condition effect between the control and HV_{long} conditions (Table 1).

Blood pH, P_{CO_2} , and $[La^-]$

Blood pH and P_{CO_2} measured immediately after the designated sprint sets gradually declined with time ($P < 0.001$, Table 2). Although blood P_{CO_2} remained lower for the 2 HV conditions than the control condition ($P < 0.001$), blood pH did not differ between the 3 conditions after the sprint sets. Pairwise comparisons showed that, in line with the P_{ETCO_2} results in Table 1, the blood P_{CO_2} values were similar between the HV_{short} and HV_{long} conditions (Table 2).

Blood $[La^-]$ significantly increased with sprint sets ($P < 0.001$, Table 2). The main effect of condition was nonsignificant; however, a significant condition $\times$ time interaction ($P = 0.034$) indicated that $[La^-]$ values, especially in later sprint sets, were greater for the HV conditions compared with the control (Table 2).

DISCUSSION

The HV interventions, HV_{short} and HV_{long}, were successfully implemented as verified by the respiratory parameters reported in Table 1. In the present study, no significant

condition effects were observed for the sprint pedaling performance (i.e., both PO_{peak} and PO_{mean}) (Figure 2). The findings illuminated that although a 30-second HV has demonstrated ergogenic performance potential (28), HV of shorter or longer duration than 30 seconds did not show consistent results (Figure 3). Where the interventions did not prove beneficial, HV_{short} may have been too short to effectively influence the excitability and contractility of the muscles during intense interval exercise. By contrast, HV_{long} may exaggerate the detrimental effects of respiratory alkalosis, (a) with impaired hemodynamics and slowed O_2 kinetics resulting in impaired recovery and (b) with reduced central motor commands (via reduced cerebral blood flow) resulting in impaired POs (3,6,7,24,32). These negative consequences could cancel out the enhancing mechanisms on performance. Impaired oxygen perfusion and diffusion are supported by the present breath-by-breath $\dot{V}O_2$ data (Figure 4B), showing that a rise in $\dot{V}O_2$ at the onset of HV was not sustained but was only transient. The abrupt drop in $\dot{V}O_2$ despite increased O_2 demands by the respiratory muscles supports the anticipated negative effects of HV on hemodynamics and O_2 kinetics (3,6,7,24,32).

Notably, the VT (and thus VE) achieved during HV_{long} was smaller than those of HV_{short} intervention (Table 1). Extended voluntary HV with near-maximum VT (i.e., forced ventilation) could trigger the Hering-Breuer reflex, which prevented overinflation or deflation of the lungs (17). Excessive hypocapnia may also inhibit the peripheral (e.g., carotid and aortic bodies) or central chemoreceptor reflex (17), leading to a fall in VT. The gradual fall in VT was indeed evident in each intervention period for the HV_{long} condition (Figure 4A). The reduced VT (hence reduced volume of HV) during HV_{long} may explain similar respiratory and blood gas profiles between the HV_{short} and HV_{long} conditions (Tables 1 and 2) despite longer breathing intervention for HV_{long}. Moreover, the discomfort sensations of HV, experienced by some subjects, may have been exaggerated particularly during HV_{long} by attempting to restore a constant VT when urged. The excessive ventilatory effort required for HV_{long} may have contributed to the nonsignificant change in performance.

The target duration of HV for ergogenic effects of repeated sprint exercises, therefore, may be around 30 seconds as reported previously (28). A limitation, however, was that this study did not include a condition of 30 seconds of HV, which could allow a direct comparison, in addition to replicating the ergogenic effects. However, inclusion of the 30-second intervention in a within-subject crossover trial would place excessive psychological stress and physical strains on the subjects. It remains plausible that the duration of HV may necessitate adjustments for different exercise types or individuals.

Hyperventilation strategies have been tested as a potential replacement for bicarbonate ingestion (metabolic alkalosis). Conflicting results have been reported among

metabolic alkalosis studies, with possible determinants being exercise types, subjects' fitness levels, or the physiological strains imposed during the exercise tests (9,32,33). We postulated that these determinants could also influence the effectiveness of HV on performance, although the process of alkalosis may be different between the 2 methods. Employing repeated sprints (previously suggested exercise modality) and trained athletes could permit imposition of heightened physiological strains. The mean data of the respiratory and blood parameters (Tables 1 and 2) may suggest that the required physiological strains were established in this study. However, inter-individual variability existed, with the maximum VE ranging 84.7–173.2 L·min⁻¹ (post-sprint set values, data not presented in Table 1), maximum [La⁻] ranging 6.89–16.47 mM, and the minimum pH ranging 7.346–7.191 under the control condition (i.e., the confounding effects of HV on these measures were disregarded). These findings show heterogeneity in the levels of physiological strains among subjects despite maximal effort, which could partly explain the inconsistency of performance change with HV_{short} or HV_{long} as depicted in Figure 3. Day-to-day differences may also contribute to the performance variability. At present, however, it remains unclear how exactly individual characteristic differences may interact with the effects of HV. Additionally, we could not establish a clear association between the sex, type of sport, or power abilities of the subjects and the effectiveness of the current HV strategies (Figure 3).

In conclusion, no significant effects were found on repeated brief pedaling sprints when HV implemented during each recovery was short (15 seconds) or long (45 seconds). The lack of ergogenic effects for HV_{long} may be ascribed to a complex interaction between the positive (augmented buffering effects) and negative effects of HV (decreased aerobic energy metabolism and exaggerated discomfort sensation of increased ventilatory work), which take place concomitantly. For the HV_{short} method, the implementation duration may have been too short to confer positive physiological effects. Hence, an optimum duration may exist, with around 30 seconds being more promising for this exercise type according to our previous study. The target duration of HV, however, may change depending on exercise types or individual characteristics and awaits further investigations.

PRACTICAL APPLICATIONS

The ergogenic effects of induced alkalosis, via either voluntary HV (respiratory alkalosis) or sodium bicarbonate ingestion (metabolic alkalosis), were mostly promising for intermittent high-intensity exercise. A remarkable feature of HV-induced alkalosis is that it can be initiated without exogenous agents and terminated instantly. This feature cannot be achieved by metabolic alkalosis induced by bicarbonate that has a relatively persistent alkalotic effect and adverse gastrointestinal effects. Hyperventilation of 30

seconds produces an ergogenic effect as seen in our previous study. When the duration of HV is too short (15 seconds) or too long (45 seconds), the ergogenic effects, however, become less promising. In addition to the implementation duration, minimizing discomfort sensation during HV may be a further consideration for ergogenic HV strategies. When in doubt, returning to spontaneous breathing is recommended because inappropriate maneuvering may result in reductions of blood flow and oxygen uptake in the periphery and the brain. These negative effects may be detrimental to performance.

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